

Technische
Universität
Dresden



Research Data Management

Digitization and Data Analytics

Ralph Müller-Pfefferkorn

Goals

- Introduction to research data management
- Basic concepts like research data life cycle, metadata or repositories
- Practical hints and examples (e.g. services offered at TU Dresden)
- No ready to use out-of-the-box system
 - Differences in disciplines
 - Personal preferences
- Think about it – Make a plan – Follow it

Agenda

1. Basic concepts
2. Metadata
3. Data organization
4. Sharing, Publishing and Finding Data

Agenda

Basic concepts

Goals

1. Understand what research data and research data management (RDM) means
2. Learn about the research data life cycle.
3. Understand why RDM is relevant for you.

What are research data?

Research data are all data that are generated or processed with a scientific method about a research object. (Jens Ludwig)

"[...] research data means any output from research projects that is not a publication."
(Rombouts & Prinčič, 2010)

What are research data?

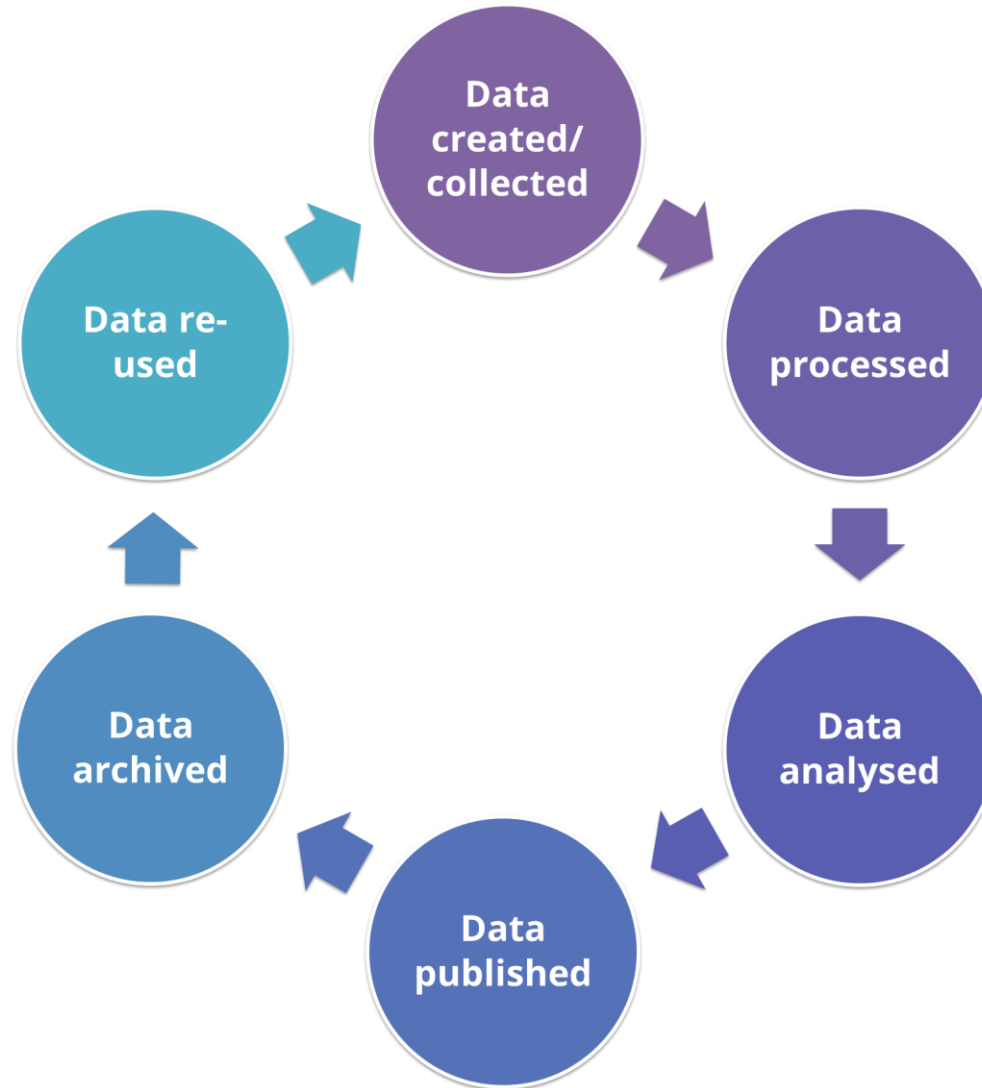
Type	Characteristics	Example
Observations	Data are collected in real-time, Usually irreplaceable	Sensor data Surveys
Experiments	Mostly produces in a laboratory environment Reproducible but expensive	Gene sequences Measurements
Simulations	Generated with models Models and metadata more important than output	Climate models Economic models
Derived data	Derived or compiled from other data Reproducible	text mining 3D models
References	Collections of usually smaller data sets Mostly published	Gene sequence database Primary text sources
Digitized data	Digitized versions of analog objects Reproducible as long as the original object is still available	Scriptures

Dominique Ritze, Kai Eckert and Magnus Pfeffer. Forschungsdaten.
In: Patrick Danowski, (Open) Linked Data in Bibliotheken; 122-138, De Gruyter Saur, Berlin, 2013

Research Data Management



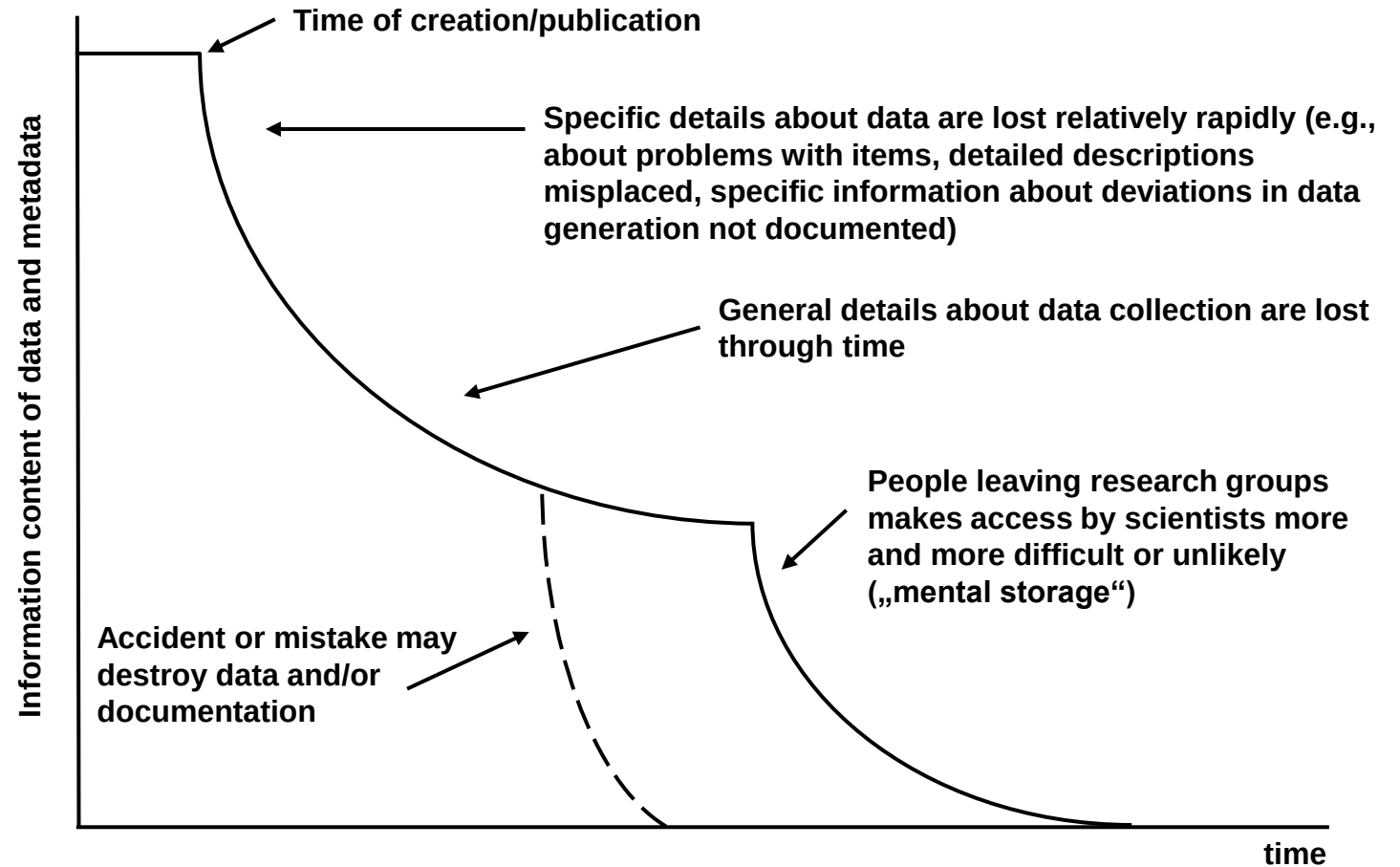
Research (Data) Lifecycle



Why research data management?

Decrease in information content of datasets over time

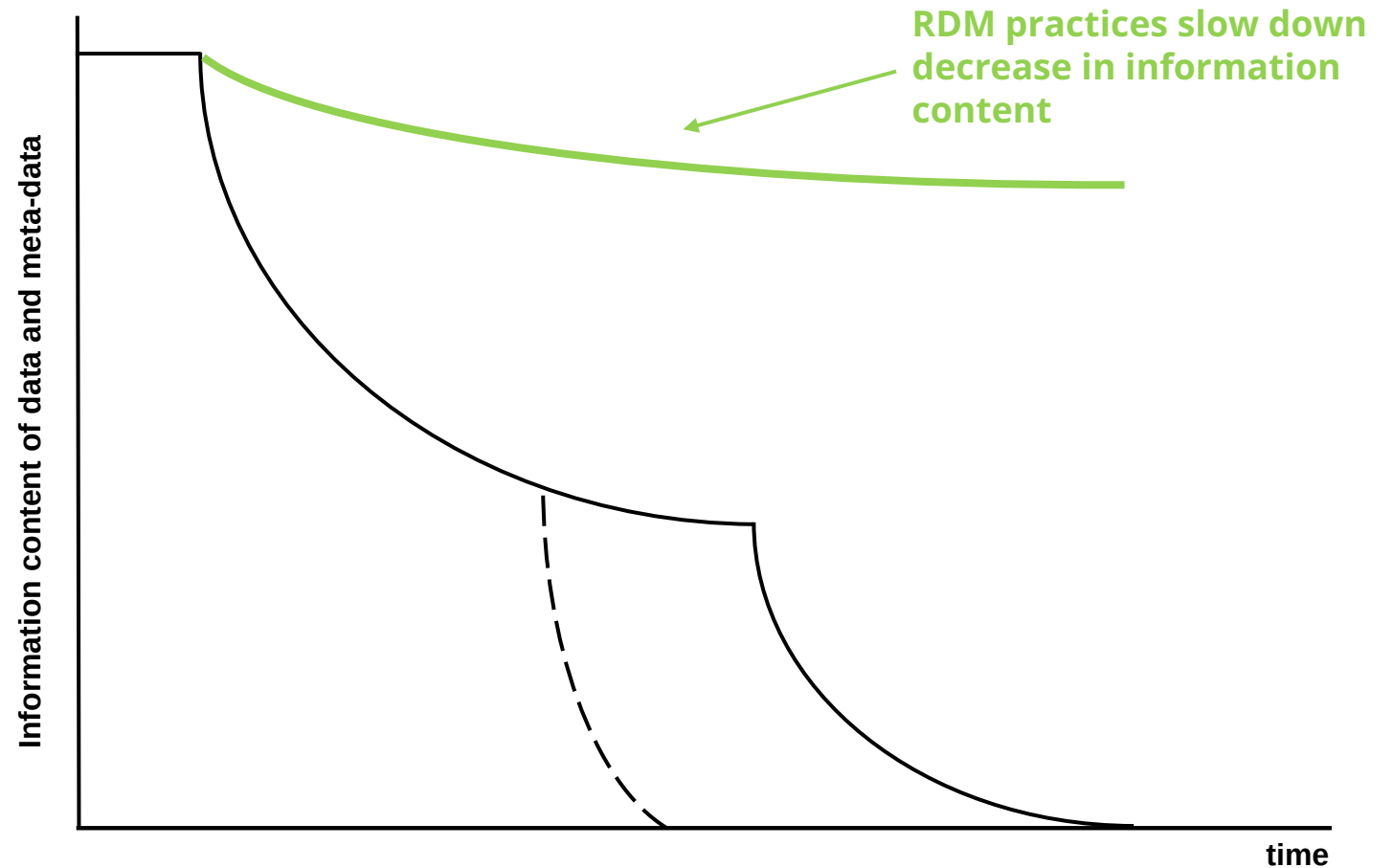
→ Information entropy



Michener, William K., et al. Nongeospatial metadata for the ecological sciences. *Ecological Applications*, 1997, 7. Jg., Nr. 1, S. 330-342.

Why research data management?

Preserving the value of data



Michener, William K., et al. Nongeospatial metadata for the ecological sciences. Ecological Applications, 1997, 7. Jg., Nr. 1, S. 330-342.

Why research data management?

For yourself!

- Keep yourself organized!
- Find your own data! Even three years after production.
- The next person you share your data with is your future self!



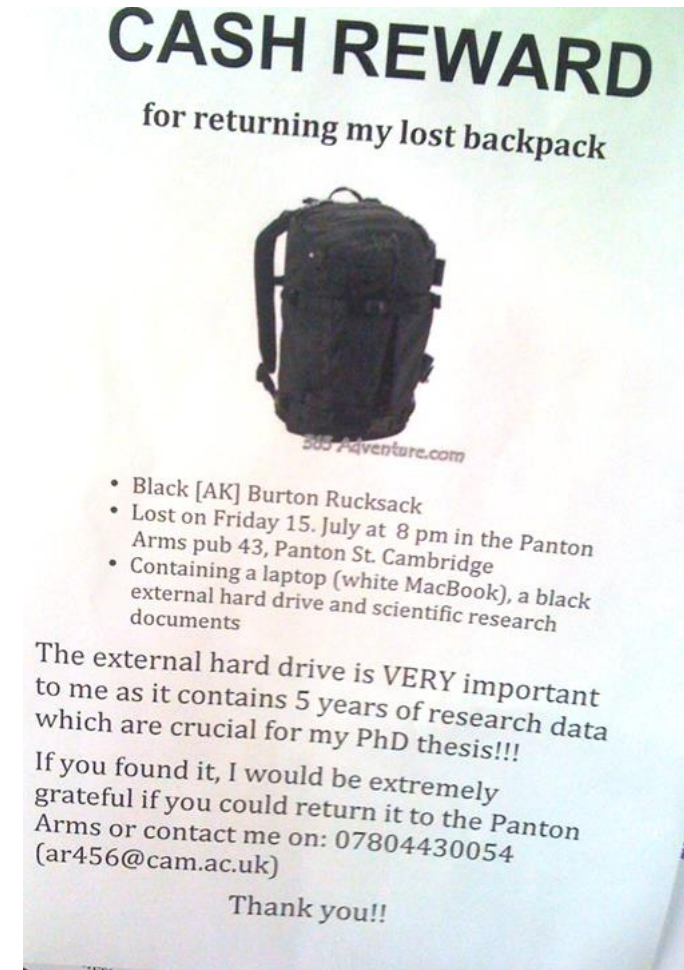
RDM is basis and prerequisite for Data Analysis!

- E.g. Machine Learning needs vast data amounts

Why research data management?

Safety / Security – Save your data!
Data are your most valuable research good!
(Beside your brain)

- Data Loss
- Natural disaster
- Facilities infrastructure failure
- Storage failure
- Server hardware/software failure
- Application software failure
- External dependencies (e.g. PKI failure)
- Format obsolescence
- Human error
- Malicious attack by human or automated agents
- Loss of staffing competencies
- Loss of institutional commitment
- Loss of financial stability
- Changes in user expectations and requirements



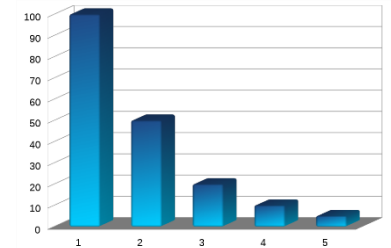
Why research data management?

Quality control, Reproducibility and Reliability of your data



Efficiency

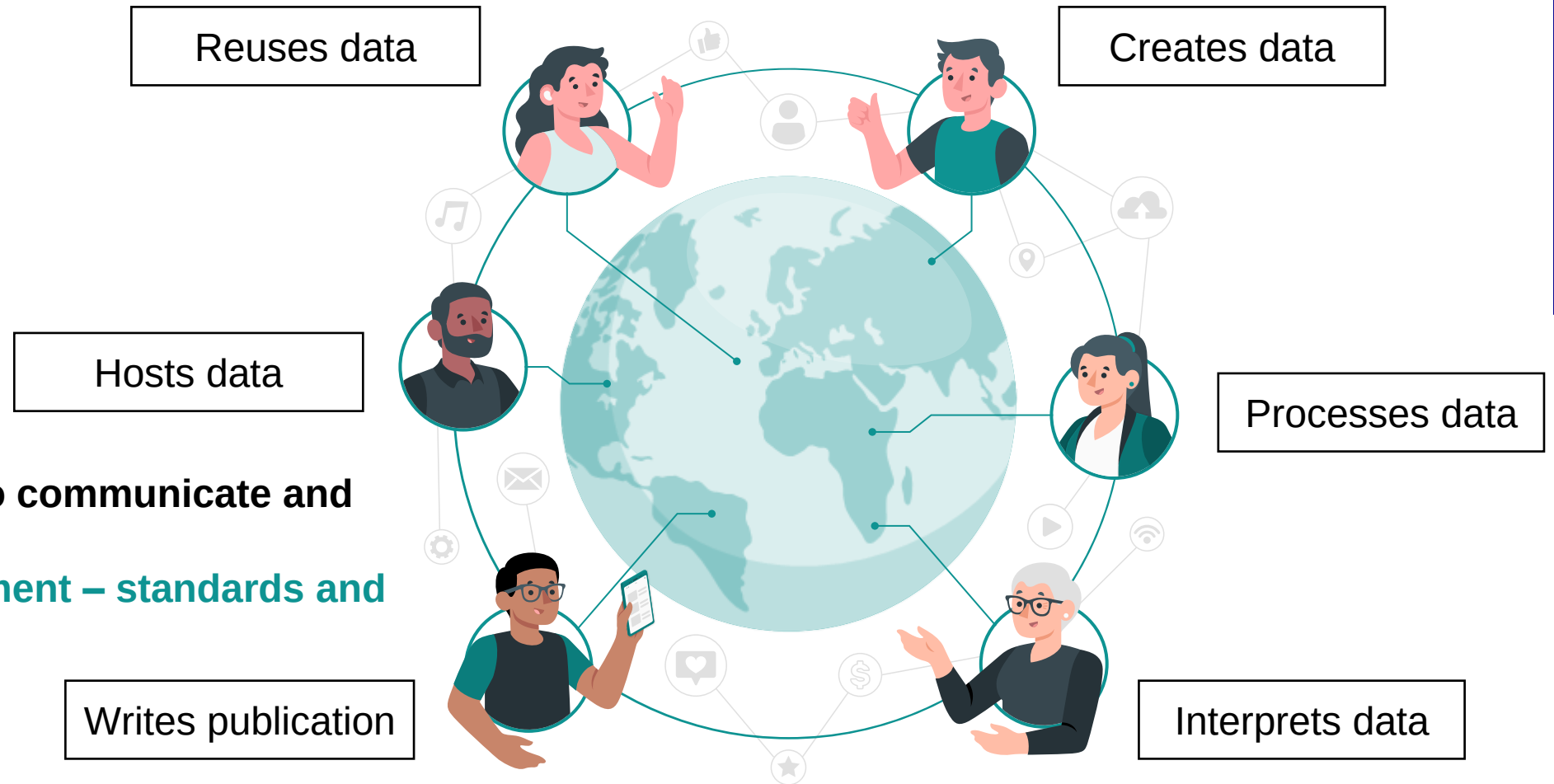
- Systematic documentation
- Consistent processes to track your research (match input and output with transformations)



Collaborative working in a group/project

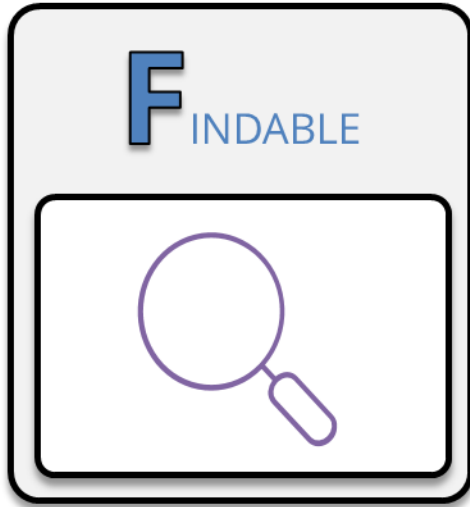


Why research data management? Communication about Data around the Data Lifecycle



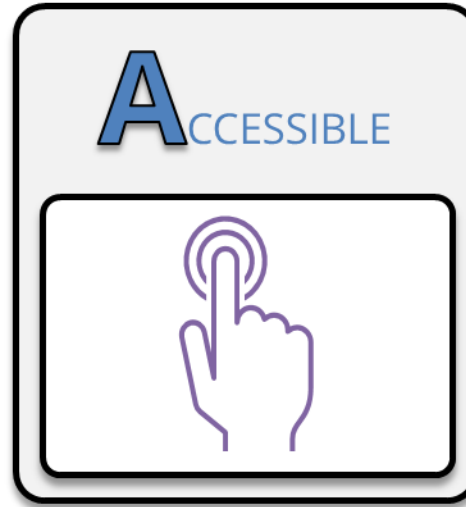
How will they be able to communicate and work together?
research data management – standards and trust

FAIR Principles



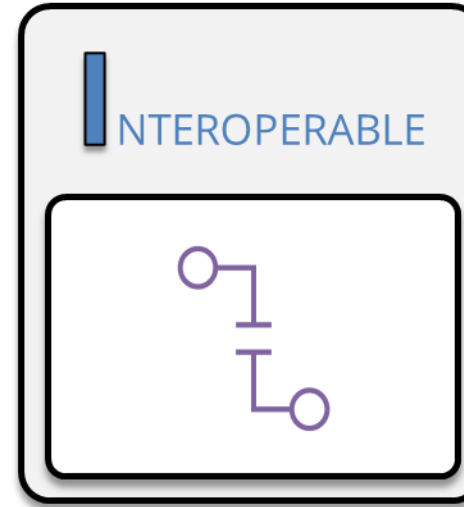
**Can I find
the data?**

**Can someone
else find the
data?**



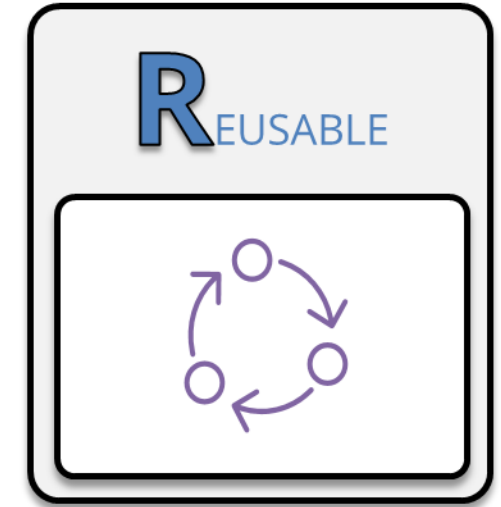
**Can I access
the data?**

**Can someone
else access the
data?**



**Can I understand
the data?**

**Can someone
else understand
the data?**



**Can I use
the data?**

**Can someone
else use the
data?**

FAIR data

Findable

- Use standardized and well-described metadata
- Use machine and human readable file and naming conventions
- Use (persistent) identifiers

Accessible

- Assign responsibility for access management
- Use platforms / repositories with standardized and well-described access protocols (e.g. REST)

→ Connect to established standards

Interoperable

- At the file level (Am I using non-proprietary file formats?)
- At the knowledge level (Do I understand the data?)
- At the connectivity level (Can I work with the data?)

Reusable

- Licenses - Do I have the right to access and use the dataset?
- Data provenance (Do I trust the origin of the data?)
- Data quality (Do I trust the content of the data?)
- Comparability (Can I use the data as a reference?)

→ Describe to increase trust

Why research data management?

Political Implications or “What funders want”

- Deutsche Forschungsgemeinschaft
- European Union
- Guidelines for the handling of research data at the TU Dresden

Political Implications or “What funders want”

1. **Responsible research goes hand in hand with a conscious and careful handling of research data over their entire life cycle**, as formulated by the DFG in its "Guidelines for the Handling of Research Data". Research data management comprises the planning, extraction, processing, documentation, publication and long-term preservation of data.
7. For the provision and subsequent re-use of the data, it is necessary to **document the data, their context of origin and the tools used**.
8. **Research data** that form the basis of a publication **should be archived and/or published in a** suitable trustworthy **data archive or repository** in the long term. They are part of the scientific reputation of the researchers.

* Link to official version (German only): (inofficial english translation)

<https://tu-dresden.de/tu-dresden/qualitaetsmanagement/ressourcen/dateien/wisprax/Leitlinien-fuer-den-Umgang-mit-Forschungsdaten-an-der-TU-Dresden.pdf?lang=de>

Political Implications or “What funders want”

3. For projects with research data, a **"Data Management Plan"** should be drawn up based on general and subject-specific guidelines (e.g. of the funding bodies or professional associations), if possible already **at the project conception** or application stage, which documents the handling of research data.
2. **Responsibility for research data management lies with the project leaders and researchers themselves.** The structural units (schools, faculties or institutes) can define binding rules for their area of responsibility.

Political Implications or “What funders want”

4. The **university supports the scientists in their research data management through a “Service Center Research Data”**, which is operated jointly by ZIH and SLUB. It offers advice on the development of concepts and support through research data management tools. This requires early contact before or at the beginning of the project.
5. The **university is equipped with a basic research data infrastructure** and thus ensures adequate storage and technical availability of digital research data.
9. For the sustainable establishment and further development of quality **research data management**, its principles and methods **should be established in teaching and training**.

Political Implications or “What funders want”

6. The University **promotes and supports free access to research data (Open Access)**
- on the one hand in the sense of good scientific practice, on the other hand because of the great importance for the optimal exploitation of research results and the quality of research. The **decision in favor of publication and its legal conditions is the responsibility of the scientists themselves**. The protection of personal data, ethical and copyright concerns, obligations towards third parties and exploitation interests should be given balanced consideration.

Summary Basic Concepts

Almost everything in research is research data as they come in many flavours.

Research data management should cover the whole research data life cycle.

You are responsible for research data management.

Plan data management before starting data taking!

FAIR data management is for the benefit of all.

Question

Was the topic of research data management already covered in your studies?

Specifically for your field of study?

Agenda

Metadata

Goals

1. Learn how to use metadata to describe your research data – for yourself and for others.
2. Learn about the different categories.

Metadata = Data Documentation

Your data **without** documentation



<https://pixabay.com/>

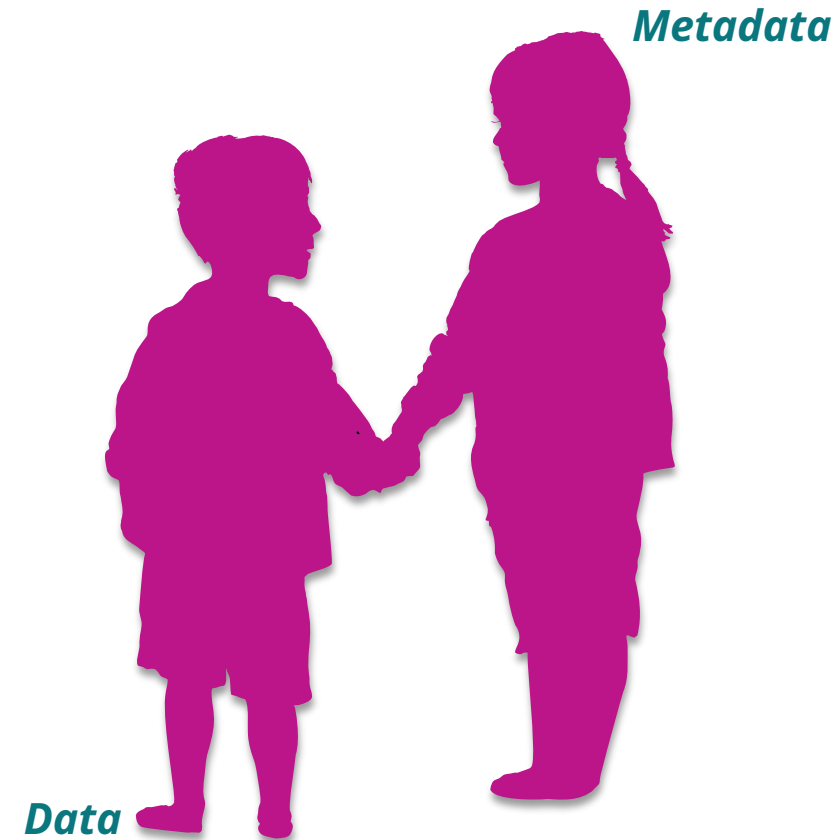
Your data **with** documentation



<https://www.unsplash.com/>

Describe to increase trust – data documentation and metadata

Data and metadata go hand in hand.



Metadata

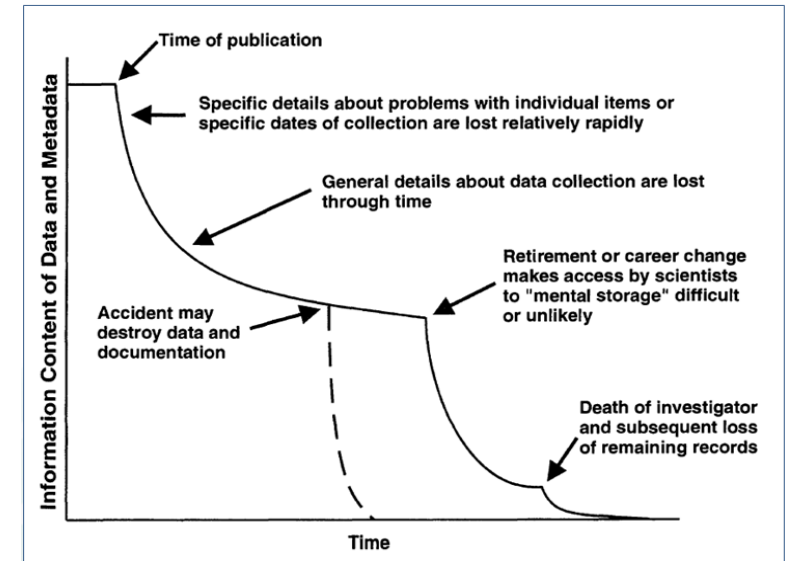
Data about data – describe knowledge about data

Why are they needed? Data are unusable without a description!

(Remember: Information entropy!)

Find data by metadata

- Where is the data set from end of May 2025 when I did this experiment with Michaels mixture at 100K temperature in the new electron microscope? Was it end of May?
- Be honest: You won't remember all the information about your data later!



MICHENER, William K., et al. Nongeospatial metadata for the ecological sciences. *Ecological Applications*, 1997, 7. Jg., Nr. 1, S. 330-342.

Metadata

Data about data – describe knowledge about data

Why are they needed?

Sharing data with others

- Vision: Other scientists understand your data without the need to contact to you.
- Otherwise: You need to talk to everybody you want to share your data with.

No long-term archiving without metadata!



Metadata and Taxonomy

Metadata provides a **description system** for “something”.

Taxonomies provide a **selection system / classification** for individual metadata fields.

Category	Description	Selection System
Form	The apple is round	List of geometrical forms
Aroma	The apple is sweet	Thesaurus of flavours
Variety	This apple variety is 'Idared'	Ontology of family „Roses“

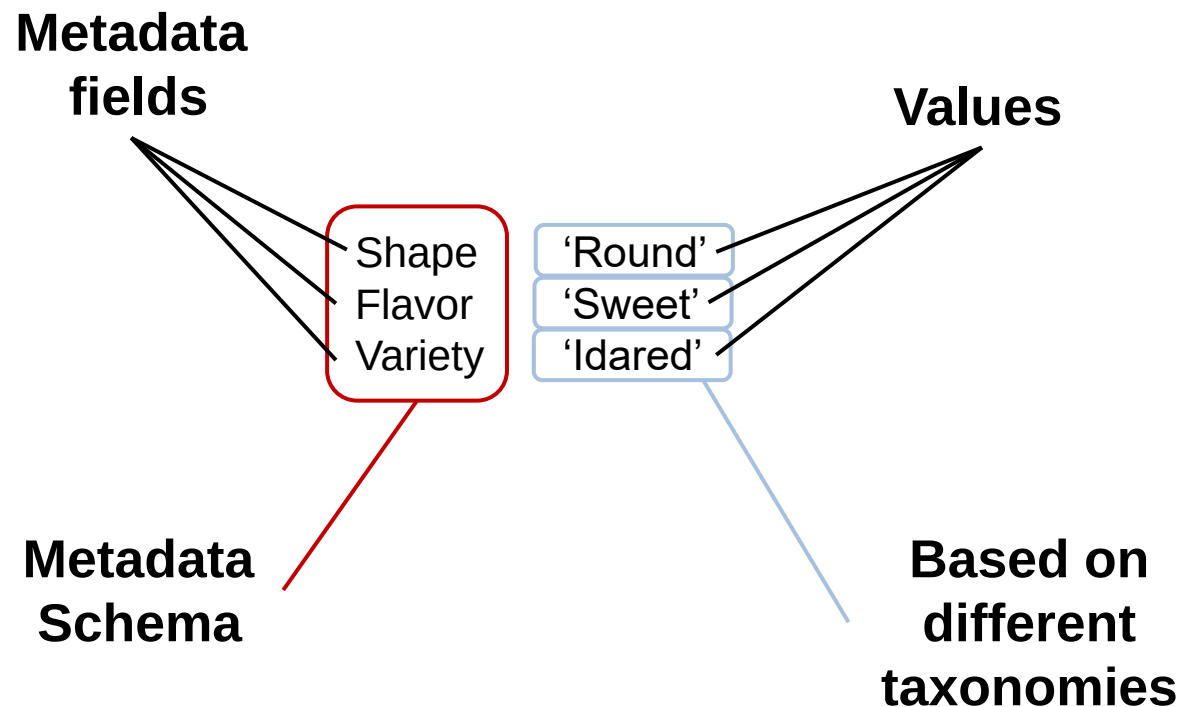


- What categories do you need to describe an apple in general?
- How would you describe this apple?
- Where do you choose your words to describe this apple?

Metadata and Taxonomy

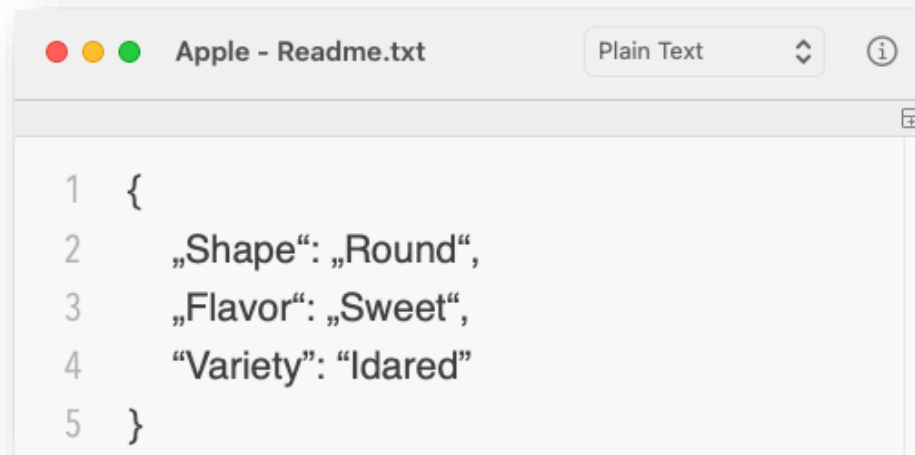
Metadata provides a **description system** for “something”.

Taxonomies provide a **selection system / classification** for individual metadata fields



Metadata and Taxonomy

A metadata file provides values for a defined metadata schema in a structured format (e.g., JSON).



```
1 {  
2   „Shape“: „Round“,  
3   „Flavor“: „Sweet“,  
4   „Variety“: „Idared“  
5 }
```



Question

Who uses metadata already in her/his daily (digital) life?

Example: Listening to music

- Spotify
- music files (like mp3) contain metadata format ID3

The screenshot shows the Spotify for Artists interface. At the top, there's a navigation bar with 'for Artists', 'Get started', 'Features', 'Resources', and 'Help'. Below this is a search bar and a breadcrumb trail: 'Home > Managing your music >'. The main heading is 'Music meta-data guidelines'. The text explains that metadata controls how songs appear on Spotify and is set by the label or distributor. It then discusses the ID3v2 format, which is a general tagging format for audio. A section titled 'Artist names' lists guidelines: adding all artist names, listing them in a separate block, considering order, keeping spelling/formatting consistent, and not allowing names that are too similar. Another section, 'Artist roles', states that roles dictate which artists are listed. The bottom part of the image shows the 'Overall tag structure' as a diagram of a container with a 10-byte header, an optional extended header, variable-length frames, optional padding, and an optional 10-byte footer.

for Artists Get started Features Resources Help

Search

Home > Managing your music >

Music meta-data guidelines

Your music's metadata controls how your songs and releases appear on Spotify. We display your music exactly as it's delivered to us, so it's important to get your metadata right.

Your metadata is set by your label or distributor before they deliver your music to us.

To fix any problems with your music's metadata, you may need to send an update to us with the correct information.

Artist names

- Add the names of all the artists and their track(s)
- List each artist's name in a separate block
- Consider the order you want to display them in
- Keep the spelling and formatting consistent across all releases to the next - include hyphens, spaces, or acronyms
- We don't allow artist names that are too similar to existing ones or Sleep Music
- To change your artist name, click here

Artist roles

- Roles dictate which artists are listed on a release

ID3v2 is a general tagging format for audio, which makes it possible to store meta data about the audio inside the audio file itself. The ID3 tag described in this document is mainly targeted at files encoded with MPEG-1/2 layer I, MPEG-1/2 layer II, MPEG-1/2 layer III and MPEG-2.5, but may work with other types of encoded audio or as a stand alone format for audio meta data.

ID3v2 is designed to be as flexible and expandable as possible to meet new meta information needs that might arise. To achieve that ID3v2 is constructed as a container for several information blocks, called frames, whose format need not be known to the software that encounters them. At the start of every frame is a unique and predefined identifier, a size descriptor that allows software to skip unknown frames and a flags field. The flags describes encoding details and if the frame should remain in the tag, should it be unknown to the software, if the file is altered.

The bitorder in ID3v2 is most significant bit first (MSB). The byteorder in multibyte numbers is most significant byte first (e.g. \$12345678 would be encoded \$12 34 56 78), also known as big endian and network byte order.

Overall tag structure:

```
+-----+
| Header (10 bytes) |
+-----+
| Extended Header |
| (variable length, OPTIONAL) |
+-----+
| Frames (variable length) |
+-----+
| Padding |
| (variable length, OPTIONAL) |
+-----+
| Footer (10 bytes, OPTIONAL) |
+-----+
```


Metadata – Categories

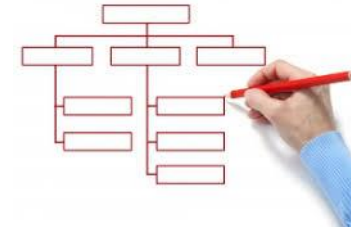
Administrative metadata

- Technical specifications and parameters, legal information
- Examples: file format, size, date and time of creation, licenses, access rights



Structural metadata

- Structure of the data and connection to other data
- Examples: table description, chapters and pages in a book



Descriptive metadata

- Describe what the data are about
- Examples: description of the experimental setup, vocabulary (predefined wording)



Metadata - How to describe data?

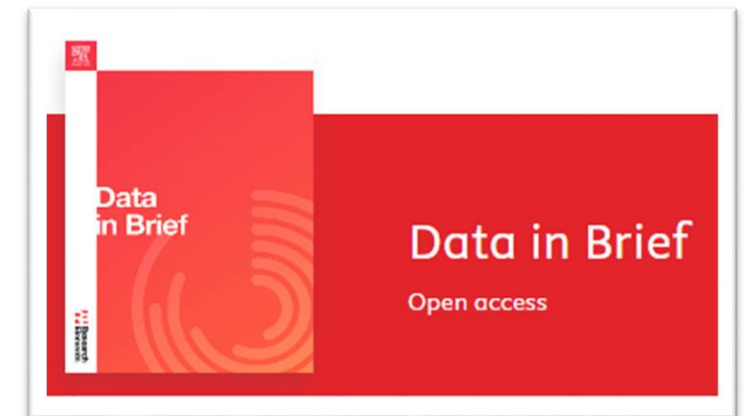
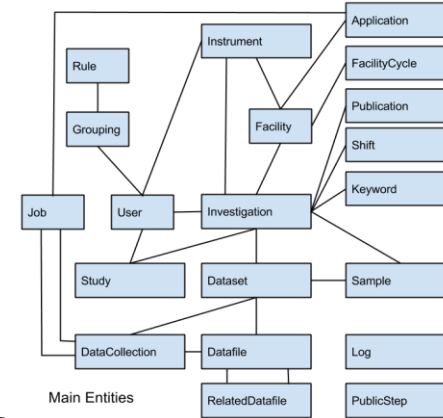
Prefer structured metadata

- Metadata schema (see later)
- Both human and machine readable
 - Human? You or your colleague
 - Machine? e.g. automatically select data sets for analysis in your analysis program

Short textual description as addition possible

- Full text indexing and search possible
- But: search results less specified → Remember the usefulness of many Google search results

Data publication – paper publication on the data (description, metadata, how to use, where to get the data ...)



Metadata Schema

Definitions of how metadata are structured and what the content should be

Metadata standards

- Standardized metadata formats
- Ease the exchange of metadata (interoperability)
- Allow machine readability

Quasi standards – no official organizational definition (like ISO or W3C), but widely used

Example for a Metadata Standard: Data Cite

ID	Property	Obligation
1	Identifier <i>(with mandatory type sub-property)</i>	M
2	Creator <i>(with optional name identifier and affiliation sub-properties)</i>	M
3	Title <i>(with optional type sub-properties)</i>	M
4	Publisher	M
5	PublicationYear	M
6	Subject <i>(with scheme sub-property)</i>	R
7	Contributor <i>(with type, name identifier, and affiliation sub-properties)</i>	R
8	Date <i>(with type sub-property)</i>	R
9	Language	O
10	ResourceType <i>(with mandatory general type description sub-property)</i>	M
11	AlternateIdentifier <i>(with type sub-property)</i>	O
12	RelatedIdentifier <i>(with type and relation type sub-properties)</i>	R
13	Size	O
14	Format	O
15	Version	O
16	Rights	O
17	Description <i>(with type sub-property)</i>	R
18	GeoLocation <i>(with point, box and polygon sub-properties)</i>	R
19	FundingReference <i>(with name, identifier, and award related sub-properties)</i>	O
20	RelatedItem <i>(with identifier, creator, title, publication year, volume, issue, number, page, publisher, edition, and contributor sub-properties)</i>	O

M = Mandatory, R = Recommend for discovery, O = Optional
<https://support.datacite.org/docs/datacite-metadata-schema-44>

Data Cite - Example

JSONRohdatenKopfzeilen

SpeichernKopierenAlle einklappenAlle ausklappenJSON durchsuchen

▼ data:

id:"10.1594/pangaea.770250"

type:"dois"

▼ attributes:

doi:"10.1594/pangaea.770250"

prefix:"10.1594"

suffix:"pangaea.770250"

identifiers:[]

alternateIdentifiers:[]

▼ creators:

▼ 0:

name:"Schumann, Kai"

givenName:"Kai"

familyName:"Schumann"

affiliation:[]

nameIdentifiers:[]

▼ 1:

name:"Völker, David"

givenName:"David"

familyName:"Völker"

▼ nameIdentifiers:

▼ 0:

schemeUri:"https://orcid.org"

nameIdentifier:"https://orcid.org/0000-0002-7237-3987"

nameIdentifierScheme:"ORCID"

affiliation:[]

▼ 2:

name:"Weinrebe, Reimer Wilhelm"

givenName:"Reimer Wilhelm"

familyName:"Weinrebe"

affiliation:[]

nameIdentifiers:[]

▼ titles:

▼ 0:

title:"Gridded results of swath bathymetric mapping of Disko Bay, Western Greenland, 2007-2008"

▼ publisher:

name:"PANGAEA"

container:{}

publicationYear:2011

subjects:[]

▼ contributors:

▼ 0:

name:"IFM-GEOMAR Leibniz-Institute Of Marine Sciences, Kiel University"

nameType:"Personal"

▼ relatedIdentifiers:	
▼ 0:	
relationType:	"IsSupplementTo"
relatedIdentifier:	"10.1016/j.quascirev.2012.02.016"
relatedIdentifierType:	"DOI"
relatedItems:	[]
▼ sizes:	
0:	"4 datasets"
▼ formats:	
0:	"application/zip"
version:	null
▼ rightsList:	
▼ 0:	
rights:	"Creative Commons Attribution 3.0 Unported"
rightsUri:	"https://creativecommons.org/licenses/by/3.0/legalcode"
schemeUri:	"https://spdx.org/licenses/"
rightsIdentifier:	"cc-by-3.0"
rightsIdentifierScheme:	"SPDX"
▼ descriptions:	
▼ 0:	
▼ description:	
description:	"A ship-based acoustic mapping campaign was conducted at the exit of Ilulissat Ice Fjord"
descriptionType:	"Abstract"
▼ 1:	
description:	
description:	"Preview maps can be downloaded at hdl:10013/epic.38398."
descriptionType:	"TechnicalInfo"
▼ 2:	
description:	
description:	"Supplement to: Schumann, Kai; Völker, David; Weinrebe, Reimer Wilhelm (2012): Acoustic m"
descriptionType:	"Other"
▼ geoLocations:	
▼ 0:	
▼ geoLocationBox:	
eastBoundLongitude:	"-50.60025200000001"
northBoundLatitude:	"69.35"
southBoundLatitude:	"69.0"
westBoundLongitude:	"-52.0"
▼ 1:	
geoLocationPlace:	"West Greenland Margin"

Metadata Schema - Disciplin specific schemas

Collections of metadata standards

- <https://rdamsc.bath.ac.uk/>
- Metadata schemas, tools, use cases
- Examples
 - Social sciences – Data Documentation Initiative (DDI)
 - Geodata – ISO 19115

Search by Discipline



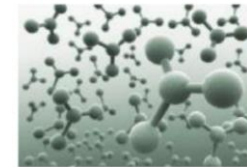
Biology



Earth Science



General Research Data



Physical Science



Social Science & Humanities

No standard available in your field?

→ Do it your own way! Don't stop documenting your data!

Question

Imagine you are a biologist.

- In a project you bought a new microscope with a camera.
- You have some cells you grew.
- You have seen some new cell features and took pictures in an experiment series.

What objects should you collect metadata for?

What do you need to collect?

Example: Disciplinary Metadata Schema - OME



OME - Open Microscopy Environment - specification for storing data on biological imaging

Includes

- image parameters (such as XYZ dimensions and pixels type)

- extensive metadata, e.g. on image acquisition, annotation, and regions of interest (ROIs)

OME community also provides software tools like

- Extraction tools to get metadata from images

- OMERO - software for visualization, management and analysis of biological microscope images using OME

Supported by hardware (microscope) and software vendors

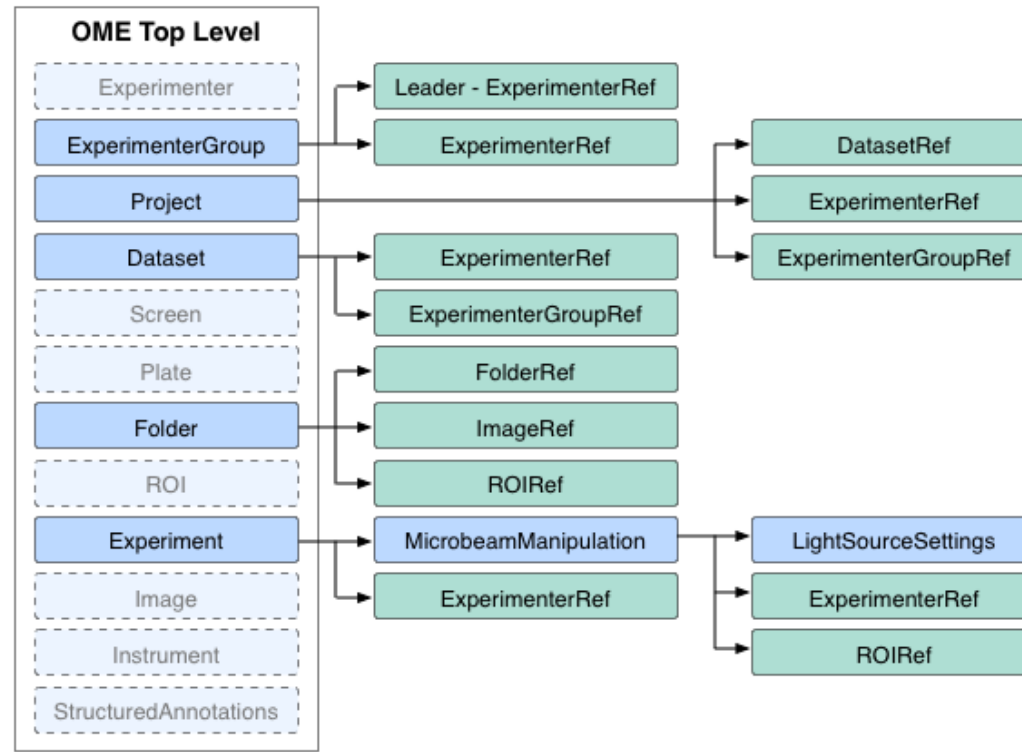
Two formats

- OME-XML – metadata in a XML file

- OME-TIFF – metadata as XML embedded in tiff-image

Example: Disciplinary Metadata Schema - OME

Organizational structures (Project, Dataset, Group, Experiment, Experimenter) of the OME Model



Which Metadata to collect?

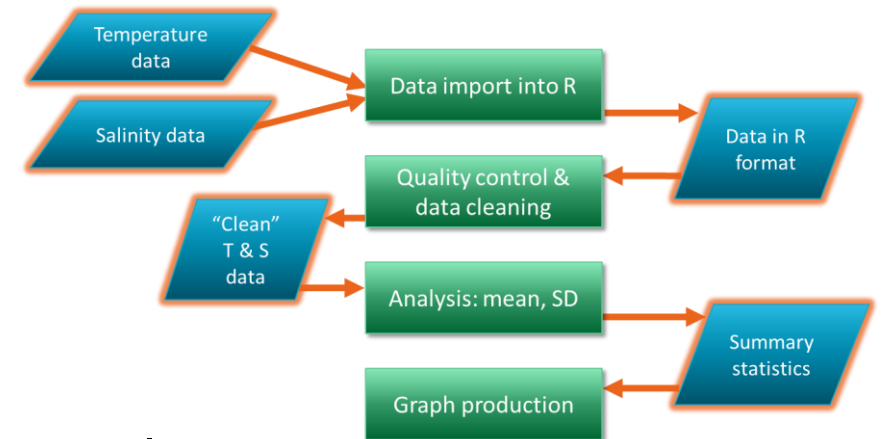
Research is a process!

- Which information objects are important in my research area?
- What must be documented?
- Where is important information generated?

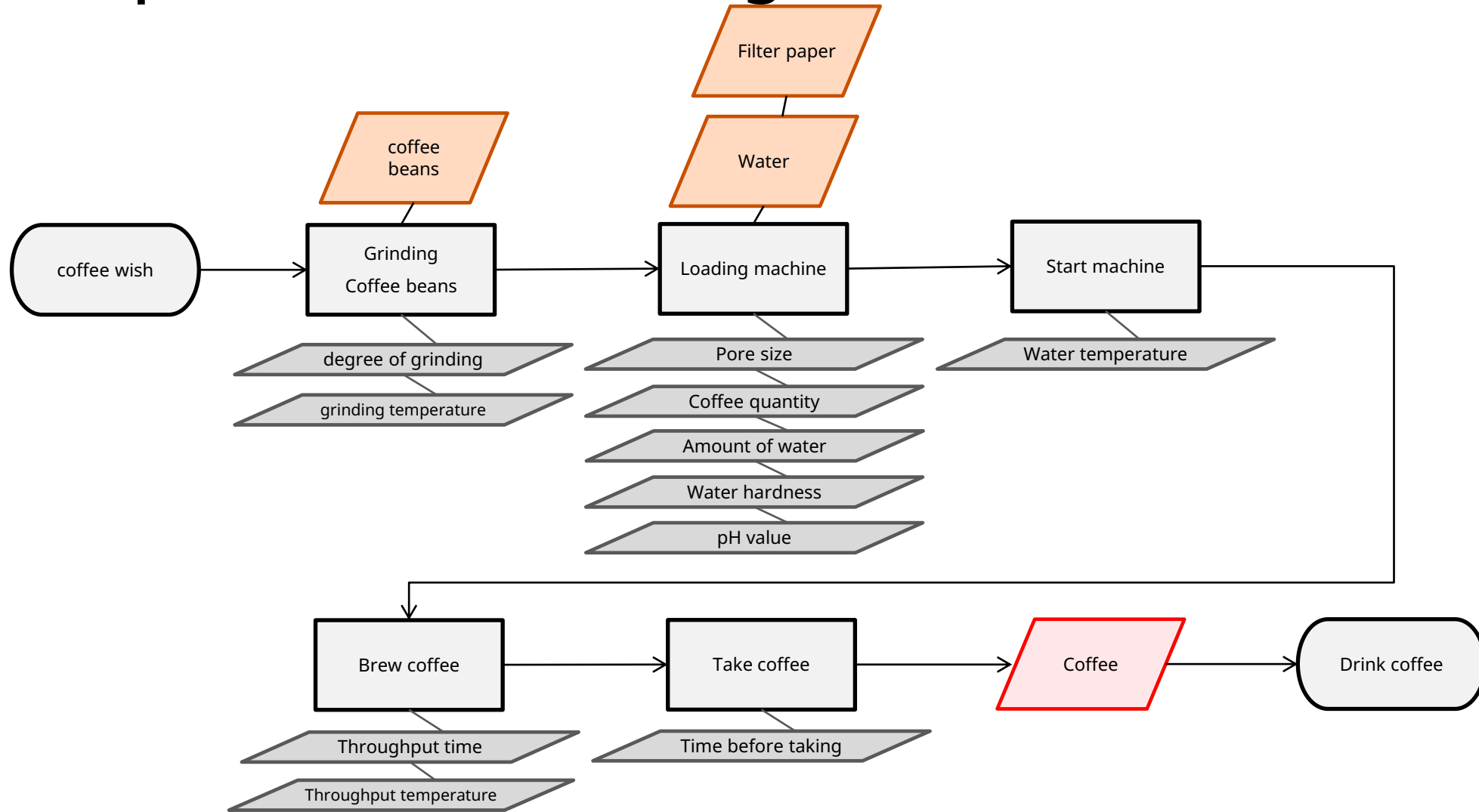
Workflows

- Can I modularize my research workflow?
- What defines a module?
- What is input/output to modules?
- What are data transformation processes (e.g. filtering, analysis)?

Visualization helps to understand documentation needs!



Example: Process Making Coffee



Other Guiding Questions for Metadata Collection

What metadata would be required to **replicate the experiment** elsewhere?

Is the experiment sufficiently explained, such that someone else could **continue the work**?

If the recordings exhibit spurious data, is the signal flow **completely reconstructable** to find the cause?

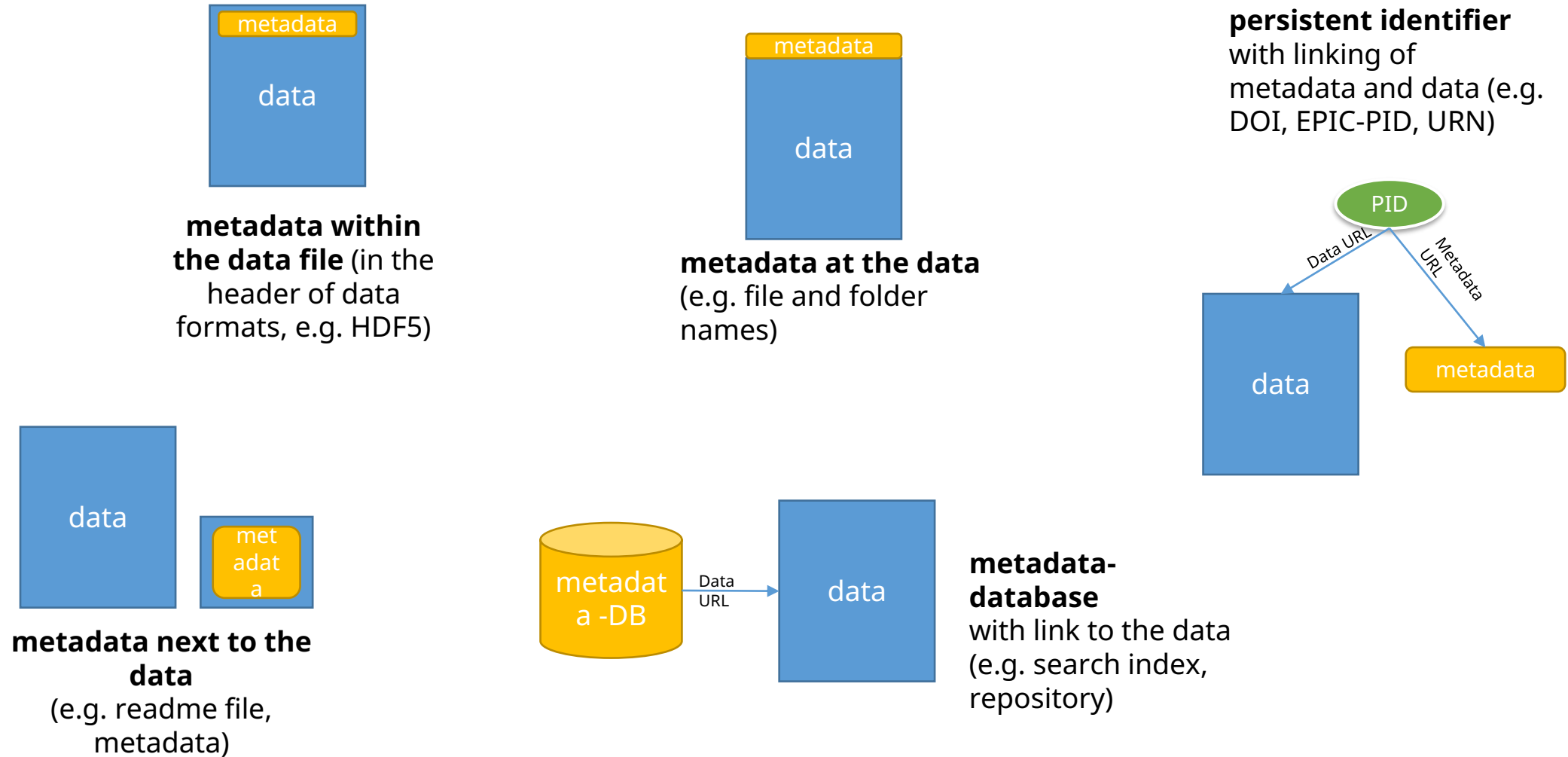
Are metadata provided which may **explain variability** (e.g., between subjects or recordings) in the recorded data?

Are metadata provided which enable **access to subsets of data to address specific scientific questions**?

Is the recorded data described in sufficient detail, such that an **external collaborator could understand** them?

Zehl, L., Jaillet, F., Stoewer, A., Grewe, J., Sobolev, A., Wachtler, T., ... Grün, S. (2016). Handling Metadata in a Neurophysiology Laboratory. Frontiers in Neuroinformatics, 10, 26. <http://doi.org/10.3389/fninf.2016.00026>

Saving Metadata – How and where?



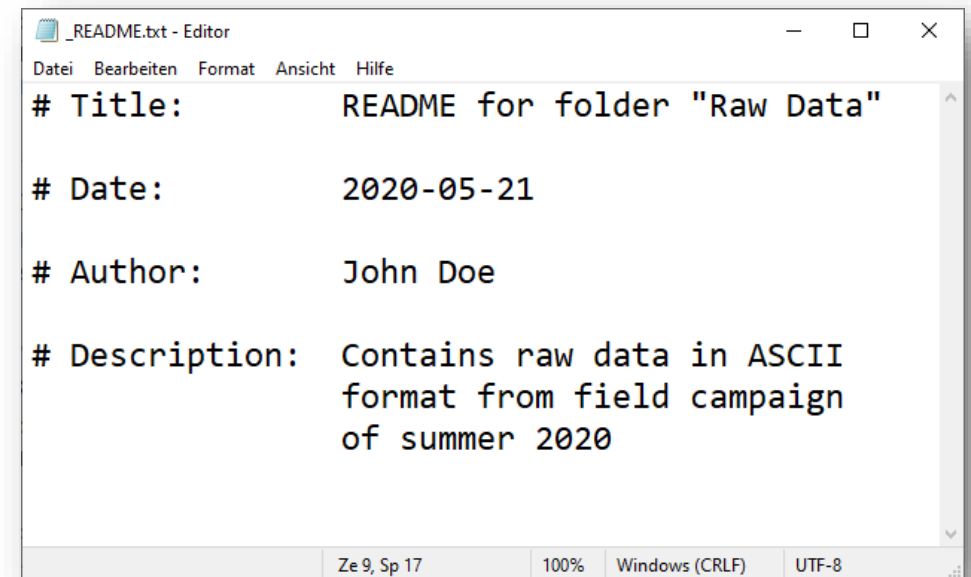
README

Easiest/fastest way to document your data:

- README.txt
- (ideally) create one in each folder
- Structured

Contents of a README.txt:

- Title
- Date
- Author
- Description of folder content
- ...



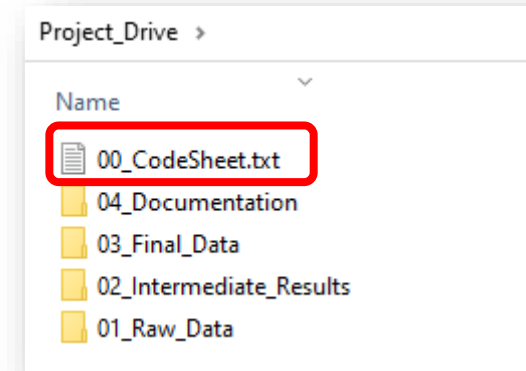
```
_README.txt - Editor
Datei  Bearbeiten  Format  Ansicht  Hilfe
# Title:      README for folder "Raw Data"
# Date:       2020-05-21
# Author:     John Doe
# Description: Contains raw data in ASCII
               format from field campaign
               of summer 2020
Ze 9, Sp 17    100%  Windows (CRLF)  UTF-8
```

<https://data.research.cornell.edu/content/readme>

Code-Sheet

Code-Sheet

- stored in top level folder
- aim: avoid misunderstandings



No	Description	Code/values
1	Gender	Male=0, Female=1
2	Age (in years)	As a continuous
3	Religion	Hindu=1, Non-Hindu =2
4	Caste	SC/ST/OBC=1, GM=2
5	Socio-economic status	As a continuous
6	Family size	≤5 =0, >5=1

source: https://www.researchgate.net/figure/Code-sheet-for-the-independent-factors_tbl1_329174118

- Possible contents of a code sheet:
- clarifying units of measurements
 - de-coding abbreviations

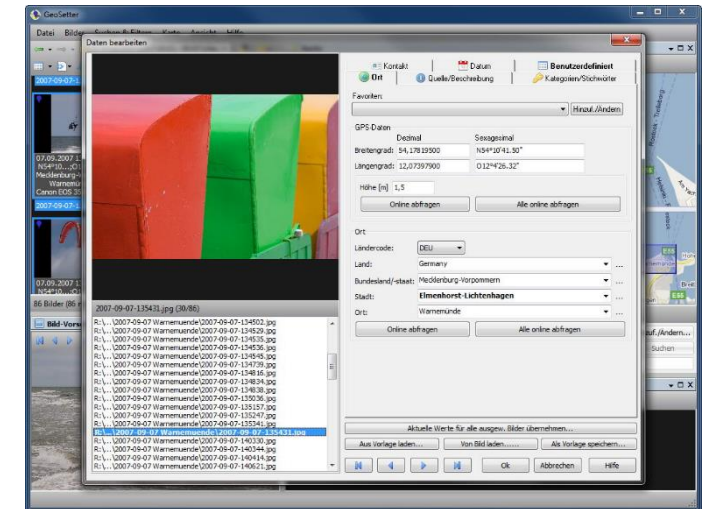
Saving Metadata – In the Data Files

Managing metadata in the data files

Tools exist for many file formats

Examples

- Geosetter – manage image data including geodata, tags ...
- Exif-tools for image metadata
- Word/LibreOffice – manage document metadata both for their own file formats and pdf



Saving Metadata – Collecting

Devices/software partly capture relevant metadata

- Camera automatically writes metadata to the image file
- Measuring instruments write metadata in a header of the measurement data
- Devices generate separate configuration files/metadata files in addition to the measurement data

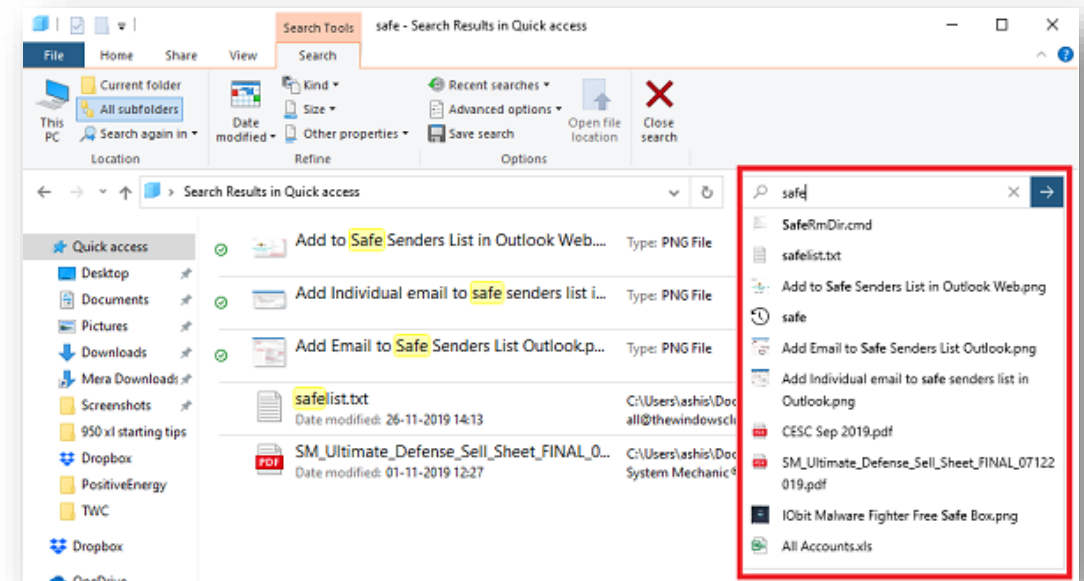
Configuring metadata acquisition in the device software

	A	B	C	D	E	F	G					
1	~Version Information											
2	VERS.	C	WLS log ASCII Standard -VERSION 2.00									
3	~Well Information Block											
4	#MNEM.UNIT	Da Description										
5	#-----	-----										
6	STRT.M		217.7796:	Starting Depth								
7	STOP.M		1267.0536:	Ending Depth								
8	NULL		-999.2500:	Absent Value								
9	COMP.	RWTH	:	Company								
10	WELL	RWTH 1	:	Well								
11	FLD.	Aachen	:	Field								
12	DATE.	02-Nov-2005	:	Log Date								
13	~Curve Information Block											
14	#MNEM.UNIes	Curve Description										
15	#-----	-----										
16	DEPT	.M	01:00	Depth								
17	GR	.GAPI	01:00	Gamma Ray counts								
18	BIT	.IN	01:00	Bit size								
19	CAL	.IN	01:00	Caliper								
20	ANI	.%	01:00	Anisotropy								
21	ANIA	.%	01:00	Average anisotropy								
22	AZSH	.deg	01:00	Corrected Tool Azimuth								
23	DAZ	.DEG	01:00	Borehole Drift Azimuth (with respect to Magnetic North)								
24	DEV	.DEG	01:00	Deviation Angle (with respect to vertical)								
25	DTSF	.US/	01:00	Fast Shear								
26	DTSS	.US/	01:00	Slow Shear								
27	FACR	.deg	01:00	Fast Shear Azimuth								
28	SACR	.deg	01:00	Slow Shear Azimuth								
29	S1IS	.%	01:00	QC Curve								
30	S1S2	.%	01:00	QC Curve								
31	S2IS	.%	01:00	QC Curve								
32	# Curve Data											
33	~A	DEPT	T	CAL	ANI	ANIA	AZSH	DAZ	DEV	DTSF	DTSS	
34	217.7796	-9		250	0	-999.2500	-999.2500	-999.2500	-999.2500	-999.2500	-999.2500	
35	217.9320	-9		250	0	-999.2500	-999.2500	-999.2500	-999.2500	-999.2500	-999.2500	
36	218.0844	-9		250	0	-999.2500	-999.2500	-999.2500	-999.2500	-999.2500	-999.2500	
37	218.2368	-9		250	0	-999.2500	-999.2500	-999.2500	-999.2500	-999.2500	-999.2500	

Searching in Metadata

Proper file naming and metadata/documentation releases full potential of desktop search tools, e.g.

- Windows Explorer Search
- Spotlight (Mac)
- Dolphin (Linux-KDE)



Source: <https://www.thewindowsclub.com/windows-10-file-explorer-search-not-working-properly>

Summary Metadata

Metadata are necessary to describe the data – both for humans and machines.

Three types of metadata: administrative, structural and descriptive metadata

Use metadata standards (if available).

Visualization and/or guiding questions help to identify metadata you should save.

Many ways to store metadata. Find the one that suits you best!

Agenda

Data Organization

Goals

1. Organizing your own data is the key.
2. Learn some basic principles.
3. An example is given for a file system based data management.

Storing Data

Short and long time

- “Living data” – data to work with
- Backup – data security
- Archiving – digital preservation

Storing Data - “Living Data”

Your PC or laptop

- Data always at your hand
- You have to care about data security – e.g. Backup



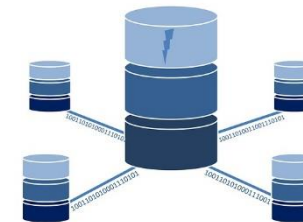
Institutional storage

- Dependence on network
- (Hopefully) somebody else cares about data security
- Sustainability (hopefully your institute will renew storage regularly)
- Ideal case: pre-defined storage structure and metadata for all institute members → make sharing of and collaborative work on data easy



Central storage

- Dependence on network
- Usually with backup and sustainable



Storing Data - “Living Data”

Central storage – Examples: Storage

ZIH HOME

- personal storage for every TU member
- You got it already with your ZIH account
- Up to 200 GB
- Automatic Backup

ZIH group storage

- storage for groups of people (e.g. projects)
- Request via ZIH Self Service Portal
- up to 100 TByte
- Automatic Backup
- Owner defines who gets access

HOME and Group drives can be directly mounted to your PC/Laptop (network drives) in Windows, Apple, Linux

Parallel file system of the High Performance Computing systems

- for very large data sets
- very fast, but no backup

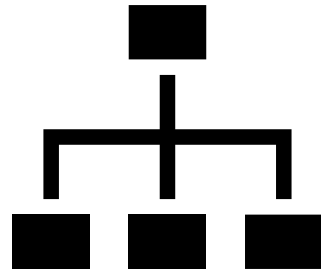
Storing Data - “Living Data”

The Cloud

- Dependence on (wide area) network
- Sometimes with backup
- (Additional costs)
- Data privacy: data are stored at the providers server
 - Who can access the data?
 - Some national laws allow (industrial) spying



File organization



Folder structure

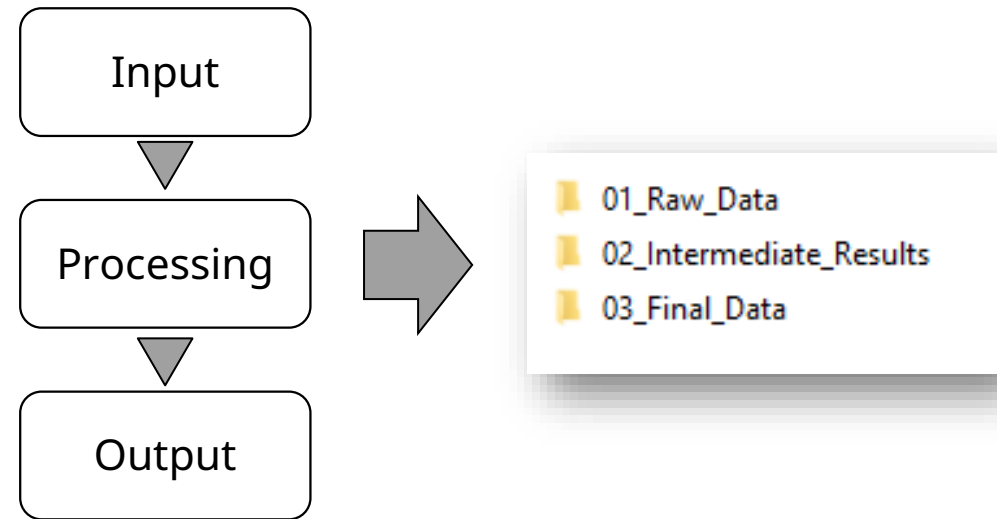


File naming

Folder Structure

Possible folder categories:

- Project name
- Time (year, exact date, ...)
- Input / Processing / Output



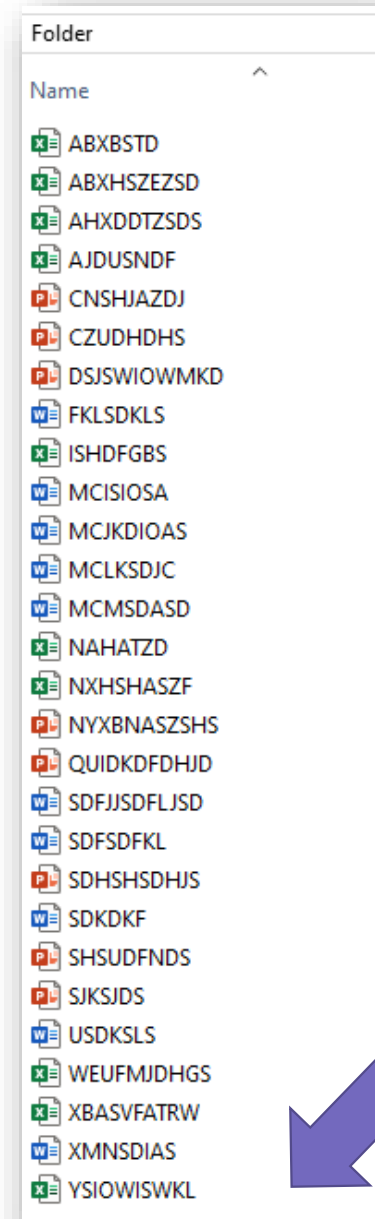
Be consistent:

follow a uniform logic!

Folder Structure

Not too big...

Rule of Seven:
once you have more than 7 files or sub-
folders in a folder, start aggregating
them into subfolders

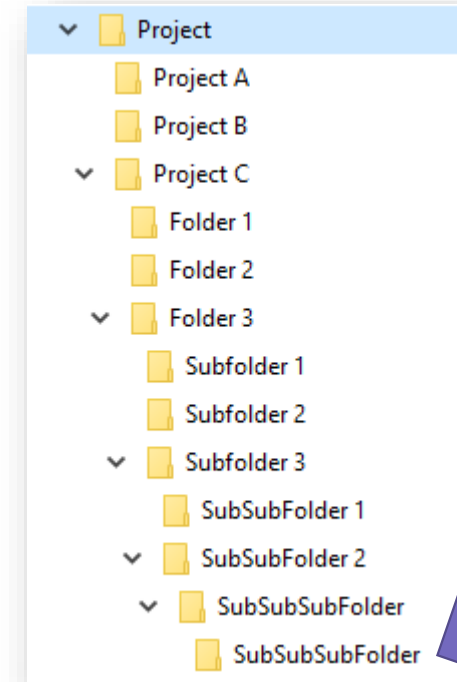


How far do you
want to scroll?

Folder Structure

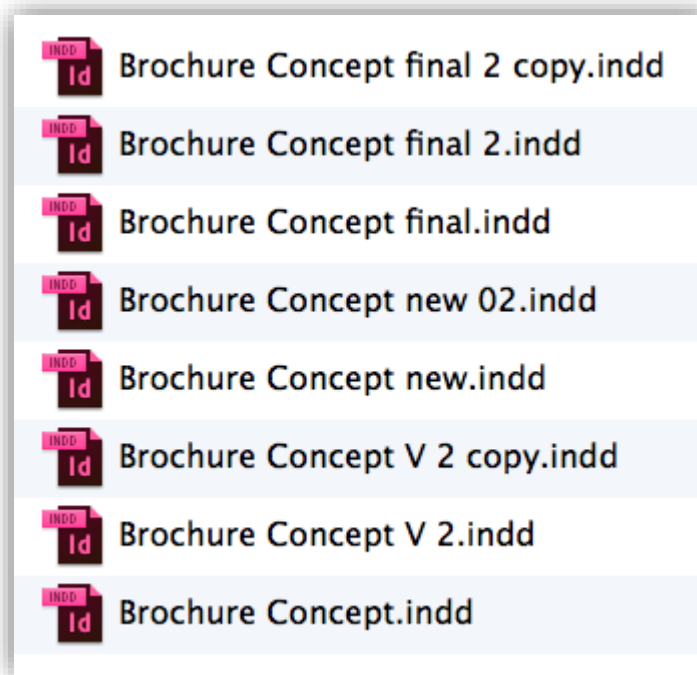
Not too deep

too many hierarchies complicate navigation



How many
clicks does it
take to get
here?

File naming



File naming

Tips for naming your files

- Use meaningful, **self-explanatory** names; avoid file names like "work", "temp"
- Settle on one convention and use it **consistently** throughout the project
- **De-code** any abbreviations or IDs in your file names in a designated code-sheet
- Naming conventions apply to **files, folders** and (!) column **headings**
- Don't use special characters in your file names such as & , * % # ; * () ! @ \$ ^ ~ ' { } [] ? < >

Example: [YYYY-MM-DD_Author_Topic]

Some tools...

Renaming files – use batch renaming tools when renaming a large number of files, e.g.

- for PC <https://www.advancedrenamer.com/> or <https://antp.be/software/renamer>
- for Mac <https://manytricks.com/namemangler/>

Removing duplicates – use duplicate finder tools to (regularly) clean up your storage, e.g.

- for PC <http://www.alldup.de/>
- for Mac <https://nektony.com/duplicate-finder-free>

Copying files – do not use Ctrl+C / Ctrl+V when copying (large) datasets; it's safer to use

- for PC Robocopy (built-in Windows utility)
- for (Mac & Linux): Rsync

Protecting Data – Backup and Archiving

Backup

- For changing data (current data set to work with)
- Several versions of a data file are kept for a (short) time
- For a restore of a version

Archiving

- For long-term preservation and re-use
- One (selected) version of a data set

Protecting Data - Backup

Make copies of (changed) files

- Restore after disk crash or virus incident
- Restore after accidentally deleting a file



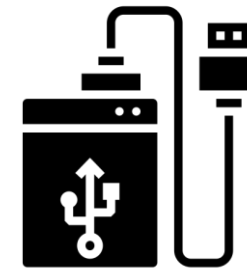
Source: Sharyn Morrow @ <https://www.flickr.com/photos/sharynmorrow/3717359715>
(CC BY-NC-ND 2.0)

3-2-1 Backup Rule

- At least 3 copies of the data
- Stored on 2 different storage media
- Of which 1 copy is physically separated



Here



Near

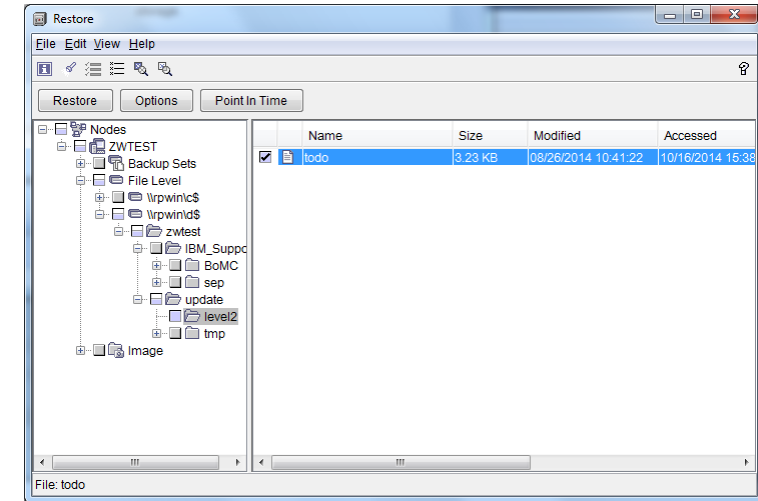


Far

Protecting Data - Backup

Example: [ZIH Backup](#)

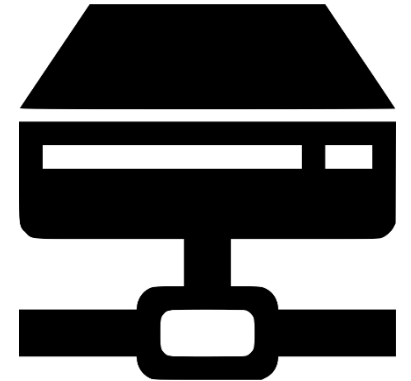
- For PCs, laptops, servers ...
- Two copies of a file on tapes
- Daily backup
- File versions are stored up to 180 days
- Graphical front end or command line tool for restore
- Restore to any place on the originating machine



TUD/ZIH Backup Services

ZIH Network drives: home drive (personal) and group drive (project)

All network drives are backed up automatically, different categories configurable, e.g.



- S1: from 8 am - 4 pm on working days one snapshot every hour, these are kept for two days; one snapshot per day for two more days; 6 more snapshots for six more weeks
- S2: one snapshot every 4 hours from 8 a.m. to 4 p.m. on work days, these are kept for two days; one snapshot per day for 12 more days; 13 snapshots each week; 6 more snapshots per month
- S3: one snapshot per day for 2 days; 12 more 1x per week
- S4: once per week for 12 weeks

Question

Who uses backup for his data?

1. At home for your PC/Laptop (e.g. for pictures, documents)
2. At your smartphone
3. At the university

Protecting Data - Archiving

Which data/metadata need to be kept after the project ends?

Five steps to follow

- Identify purposes that the data could fulfil (e.g. verification, further analysis, re-use)
- Must it be kept? (e.g. as evidence or for legal reasons)
- Should it be kept? (e.g. for its potential value)
- Consider costs – do benefits outweigh cost?
- Make a decision e.g. using the table from here:

www.dcc.ac.uk/resources/how-guides/five-steps-decide-what-data-keep



Preparing data for archiving

Do not leave data preparation for last

Establish data cleaning routines, e.g.

- 5-10 min every Friday afternoon
- group events („Editathon“)

Constantly focus on decreasing amount of ROT data (= **R**edundant, **O**bssolete, or **T**rivial)*

for further information please check Source: Gorman & Aiken - The case for the chief data officer (2013)

Protecting Data – Archiving – File Formats

Data formats change

→ Older formats no longer readable

Not all formats appropriate for archiving

- Recommendation: use open standards instead of proprietary formats
- Archiving services usually recommend formats



Protecting Data – Archiving – File Formats

Text

- PDF/A (*.pdf)
- Unformatted text (*.txt, *.c, *.cpp, *.m, usw.)
 - ASCII, UTF-8, or UTF-16
- XML

Images

- TIFF (*.tif) uncompressed, if possible TIFF 6.0, Baseline TIFF
- PNG (*.png)
- JPEG2000 (*.jpg, *.jpeg)
- SVG) (*.svg, *.svgz) without Javascript binding

Spreadsheets

- Comma/semicolon/tab separated values in text files (*.csv)

Multimedia

- WAV (*.wav) uncompressed
- Motion JPEG 2000 (ISO/ IEC 15444-4) (*.mj2)
- AVI (*.avi) uncompressed, motion JPEG
- QuickTime Movie (*.mov) uncompressed, motion JPEG

<https://www.forschungsdaten.info/themen/bewahren-und-nachnutzen/formate-erhalten/>

Protecting Data – Archiving - Metadata

No archiving without metadata!



See before!

Protecting Data – Archiving at TUD

OpARA – Open Access Repository and Archive

- Archiving and publishing of research data
- See later

Summary Data Organization

Find your way to organize your data!

- Where to store? How to structure?
- Backup und archiving
- Follow your concept!

Think of backup and archiving even at the beginning.

Agenda

Sharing, Publishing and Finding Data

Goals

1. Understand the different forms of data sharing.
2. Learn how to publish research data.
3. Understand that you can find and reuse the research data of others.

Sharing Data

Essential for collaborative research

Various forms of sharing/cooperation

- Allow others to directly access/use your data
- Cooperative work within a group
- Publishing data to the world

THE MORE WE SHARE,
THE MORE WE HAVE.

- LEONARD NIMOY

Sharing Data – Direct Access

Allow others to access/use your data

Usually for a selected set of people – Project members, Students ...

Access and access rights have to be given

Again: Metadata needed to understand the data

ZIH group storage

- Central directory
- Group can be arbitrary defined
- Anyone with a ZIH login (including guest logins)
- All group members can mount the directory to their PC/Laptop

ZIH Datashare

- Share a file or directory (e.g. via a link sent to somebody)
- Password protection possible
- Browser interface, clients for Windows, Linux, Mac, Android, iOS
- Easy and intuitive to use

Sharing Data – Cooperative Work



[Sharepoint at ZIH](#)

Web browser based collaboration platform for every member of TUD (via ZIH login)

Project sites

Document/data management; collaborative editing of documents (Word, Excel, Powerpoint), even in Browser; basic metadata (author, date ..) automatically created - additional metadata fields can be added; Task management; Wiki, Notes, Calendar ...

Publishing Data

Highest level of data sharing

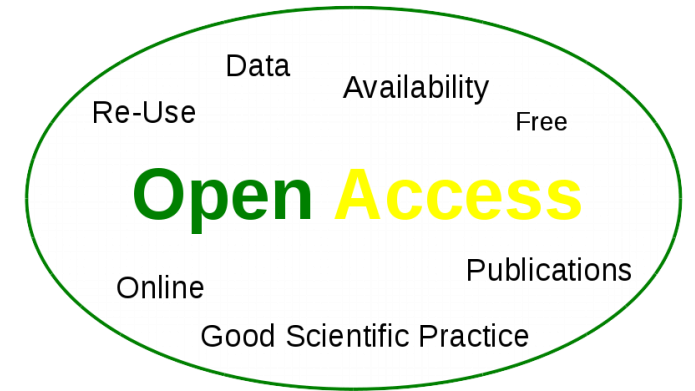
Why publishing data?

- Data production as novel scientific activity - Citation of data
- Visibility of your research
- Your research is publicly funded, make your data available for the public.

How to publish data?

- As individual data sets in a repository
- In a data journal with description
- As supplement of a paper, e.g. cooperation of Elsevier & Pangea

Some publishers have requirements on data publishing e.g. Nature, PloS



Publishing Data

How to cite data?

Like other citations:

Data sets get a Digital Object Identifier (DOI)

- Persistent and unique link/URL
- Comparable to ISBN for books
- Used both for papers and data
- Points to a web page with the data and metadata
- Example: doi.org/10.25532/OPARA-1038

M. Trommler, “Airborne Laser Scanning, Elevation Models and True Orthophoto Mosaic of the Saxon–Bohemian Switzerland National Park Region from 2005.” Technische Universität Dresden, Apr. 27, 2026, doi: 10.25532/OPARA-1083.

Publishing Data - Repositories

Repository (aka Data Repository or Digital Data Repository): Service for archiving and publishing digital data

- Standardized interfaces (e.g. via web page)
- Searchability
- Retrieval of the data
- Processes for data preservation and data curation

Types

- Disciplinary/domain-specific
- Generic
- Institutional
- Media-specific (texts, images, movies, etc.)

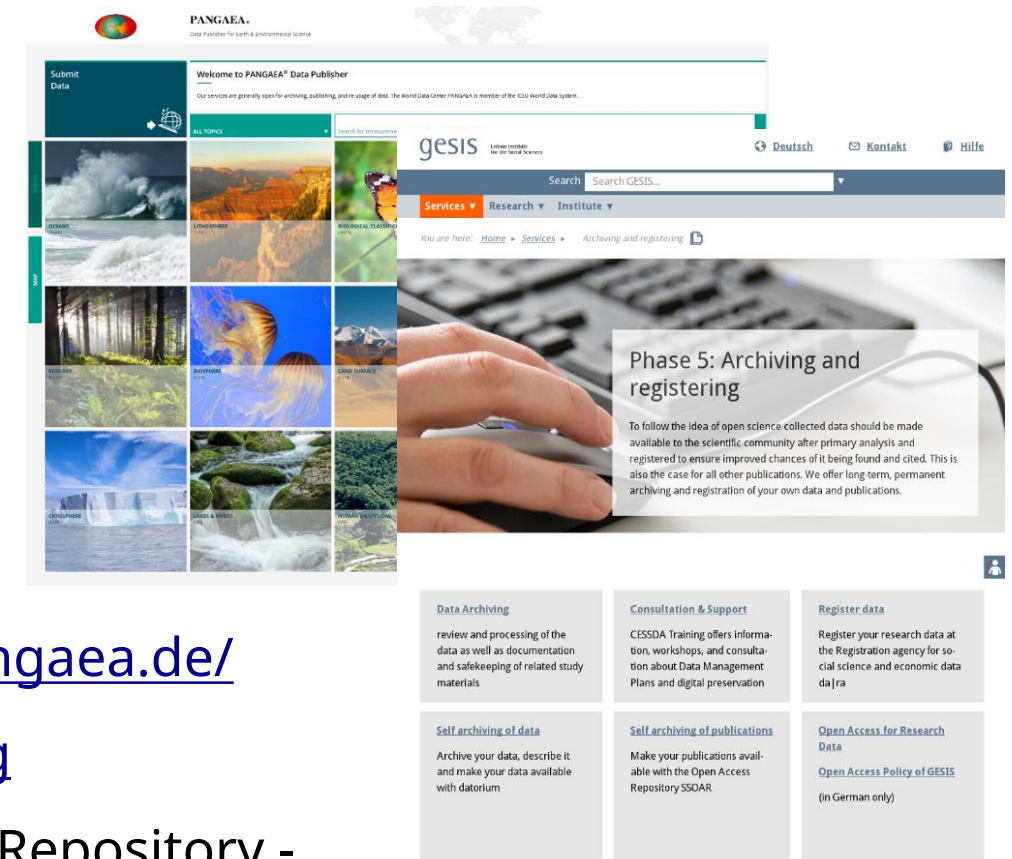
Publishing Data - Repositories

Disciplinary/domain repositories

- For one scientific discipline/domain
- Usually fixed metadata schema
- A central place to look for data in a domain
- Use it if there is one for you!

Examples

- Pangea – Geodata repository - <https://www.pangaea.de/>
- GESIS – Social sciences – <https://www.gesis.org>
- Nomad Repository - Novel Materials Discovery Repository - <http://nomad-repository.eu/cms/>



Publishing Data - Repositories

General/institutional repositories

- e.g., institutional repositories for one institution like a university
- Open for many disciplinary data and metadata schemas
- For domains that cannot afford a domain repository
- Disadvantage: Often no support of discipline-specific data and metadata

Examples

- OPARA – the TU Dresden repository - <https://opara.zih.tu-dresden.de/>
- Zenodo (CERN) - <https://zenodo.org/>

Publishing Data - OpARA

Archive, publish and find research data

- For TU Dresden, TUBA Freiberg, HTW Dresden, Universität Leipzig
- Triple redundant and spatially separated storage of the data in tape archives
- Data ingest via web browser or command line tool



opara.zih.tu-dresden.de

The screenshot shows the OPARA Research Data Repository website. The header includes the OPARA logo, navigation links for 'All of OPARA' and 'Statistics', and a 'Log In' button. The main heading is 'Research Data Repository of Saxon Universities'. Below this, a paragraph states: 'OPARA is the Open Access Repository and Archive for Research Data of Saxon Universities.' Another paragraph explains that researchers can publish or archive data here to meet funding agency requirements. A note mentions that the service was recently upgraded and that previously stored data will be migrated to a new version. A search bar is located below the text. The 'Communities in OPARA' section lists four institutions with their logos: Hochschule für Technik und Wirtschaft Dresden (HTW), Technische Universität Bergakademie Freiberg (TUBAF), Technische Universität Dresden, and Universität Leipzig. The 'Recent Submissions' section shows a submission titled 'Multi-Domain Optimization for Networked Control Systems' from Technische Universität Dresden, dated 2026-04-30, by Pallewaththe Kankanamge, Hasal Dinusankha Kulasekara, and Soler Perez Oliva. It describes the repository as containing experimental artifacts for a paper on multi-domain model-driven optimization for networked control systems.

Publishing Data - OpARA

Services

- **Free archiving** of research data for an initial period of **ten years** for scientists of the participating institutions
- Unlimited storage for published data
- Optional **publication**
- Various models: Open Access, Embargo, Limited access
- Assignment of **DOI** for the permanent citation of data
- **Quality assurance** through double review
- Local service provider and local contact

Finding Data

Interdisciplinary Databases and Search Engines

Database / Search Engine	Link	Notes
DataCite	https://search.datacite.org/	global non-profit organization that manages DOIs
Base	https://www.base-search.net/	Universität Bielefeld
EuDAT	http://b2find.eudat.eu	EU funded initiative
OpenAIRE	https://explore.openaire.eu/	EU funded initiative
CrossRef	https://search.crossref.org/	Initiative founded jointly by publishers

Summary Sharing, Publishing and Finding Data

Are your data valuable for others to? Share or publish it.

Use research data repositories.

You can find research data (or repositories with data) of others in the internet.

Support at TUD - Service Center Research Data

“Kontaktstelle Forschungsdaten”

Consulting and supporting scientists

Organization of research data management

- Data management plans
- Metadata
- Tools and services
- Archiving
- Publishing data
- Legal issues ...



Contact

eMail: Kontaktstelle-Forschungsdaten@tu-dresden.de

Web page:

<https://tu-dresden.de/forschung/services-fuer-forschende/kontaktstelle-forschungsdaten>

Final Summary

Research Data Management is the base for Data Analytics!

Find your own way in research data management!

Find the right balance between effort and benefits!

Plan it (before)!

Document it!

Use help (support, tools, metadata schemas ...)!